Understanding Audio and Speech with Language Models

NLP Summer School 2026

Kempelen Institute of Intelligent Technologies

Language is not only text

“Human language is firstly spoken and only secondarily written.”

Topics for today

- Why is audio processing useful?
- Benchmarks
- Overview of speech-to-text models
- How does a language model read audio?
- Capabilities of current open models
- Audio data sources

But first, open demos and download models!

- Speech-to-text Colab Demo
- Text-to-Speech Colab Demo

tinyurl.com/kinitnlp

What can we do with audio models?

- Speech to Text
- Text to Speech
- Voice to Voice
- Media generation
- Audio classification

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- Text to Speech
- Voice to Voice
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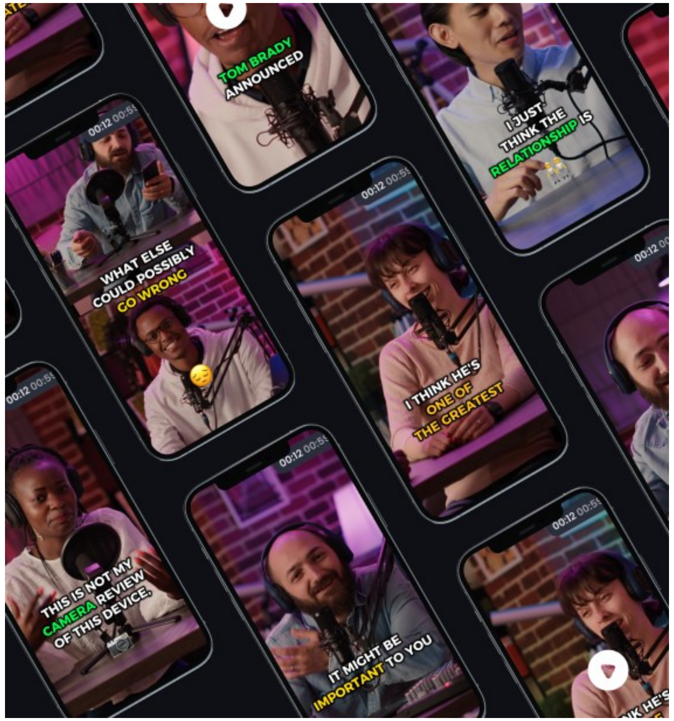
This is our current focus

What exactly we do with audio models in KInIT?

- KInIT fine-tuned ASR models on Hugging Face
- Slovak speech-to-text benchmark
- Slovak language models

Audio model applications

- Automated transcription: subtitles, notes, medical records



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- **Talk to machines: virtual assistants, agents, smart devices, robots**

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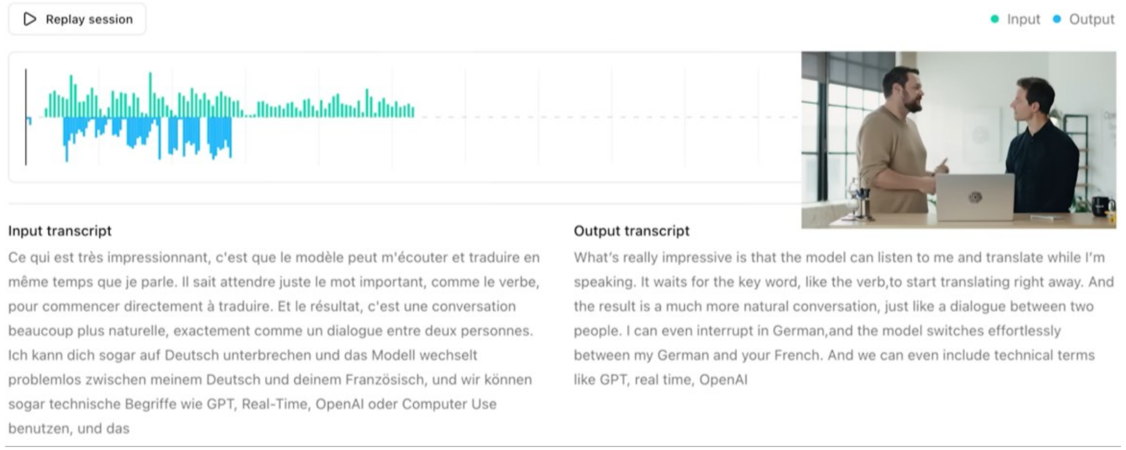
- **Real-time speech translation**



Audio model applications

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- **Music and sound effect generation, automated editing**

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- Music and sound effect generation, automated editing
- **Machine monitoring and predictive maintenance**

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We can do unexpected things with audio models

Speech-to-text Models

Overview & benchmarks

Leaderboard | Multilingual | Long-form | Private data | Pareto | About

Select columns to display:

- Toggling datasets will change the Average WER. Default datasets: AMI-Cleaned, Earnings22, Gigaspeech-Cleaned, LS Clean, LS Other, SPGISpeech, Voxpopuli-AA-Cleaned, Private (scripted), Private (conversational).
- **Rank Δ** shows how the ranking changes compared to the average over the default datasets.
- **Private (scripted)** and **Private (conversational)** columns show aggregated WER on private datasets from Appen Inc. and DataoceanAI (see the 'Private data' tab for more info).

Columns

- [x] RTFx
- [ ] License
- [ ] Size (B)
- [ ] # Languages
- [ ] Encoder
- [ ] Decoder
- [x] AMI-Cleaned
- [x] Earnings22
- [x] Gigaspeech-Cleaned
- [x] LS Clean
- [x] LS Other
- [x] SPGISpeech
- [x] Voxpopuli-AA-Cleaned
- [ ] AMI
- [ ] Gigaspeech
- [ ] Voxpopuli
- [x] Private (scripted)
- [x] Private (conversational)

Rank	Rank Δ	model	Average WER …	RTFx	AMI-Clean…	Earnings22	Gigaspeech-Clean…	LS Cle…	LS Oth…	SPGISpeech	Voxpopuli-AA-Cleaned	Private (scripte…	Private
1	–	modulate/vfast	5.37	NA	6.72	8.09	7.25	0.92	1.88	2.41	3.74	2.68	14.62
2	–	microsoft/azure-speech-06-2026	5.39	NA	6.95	7.52	7.26	1.19	2.49	2.31	3.87	2.27	14.63
3	–	reson8/resonant-1	5.54	NA	8.22	8.79	7.39	0.92	2.08	2.77	2.19	2.96	14.54
4	–	zoom/scribe_v1	5.58	NA	6.94	9.17	7.87	1.12	2.08	1.43	3.95	2.81	14.87
5	–	AutoArk-AI/ARK-ASR-3B	5.59	484.29	7.89	8.23	6.94	1.03	2.35	2.46	3.55	3.11	14.78
6	–	reson8/resonant-1-flash	5.59	NA	8.21	8.96	7.39	0.94	2.13	2.8	2.19	2.99	14.67
7	–	HojoAI/Hojo-ASR-V1	5.6	73.8	7.53	8.15	6.14	1.3	3.24	1.74	3.11	3.99	15.21
8	–	elevenlabs/scribe_v2	5.7	NA	8.2	8.9	7.64	1.05	2.32	3.01	1.41	3	15.76
9	–	Qwen/Qwen3-ASR-1.7B-hf	5.74	796.19	8.3	9.73	7.23	1.26	2.93	2.57	2.85	2.87	13.9
10	–	Qwen/Qwen3-ASR-1.7B	5.78	394.13	8.31	9.88	7.22	1.24	2.92	2.58	3.01	2.9	13.95
11	–	bosonai/higgs-audio-v3-stt	5.83	110.12	7.19	8.26	7.45	1.07	2.6	2.07	3.73	3.74	16.39
12	–	assemblyai/universal-3-5-pro	5.84	NA	10.63	10.02	7.62	1.1	2.1	1.81	1.92	2.66	14.71

Multilingual ASR Evaluation

Select languages to include in the ranking. The **Average WER** is the mean over the selected languages, and a model is ranked only if it supports **all** of them (apples-to-apples over the same set).

Models that cover only *some* of the selected languages aren't dropped — they appear below the ranked models as unranked rows (Average WER `NA`), with `NA` in the languages they don't cover. The **Coverage** column (e.g. `3/5`) shows how many of the selected languages each model supports.

Use **Language dataset breakdown** to break down the results for a single language's per-dataset results; the other languages are hidden when specifying a single language.

Toggle GB **English** (off by default) to include English alongside the other languages. Its score is the average over the same datasets as the main leaderboard: AMI-Cleaned, Earnings22, Gigaspeech-Cleaned, LS Clean, LS Other, SPGISpeech, Voxpopuli-AA-Cleaned, Private (scripted), Private (conversational).

Languages

- [x] DE German
- [x] FR French
- [x] IT Italian
- [x] ES Spanish
- [x] PT Portuguese
- [ ] GB English

Language dataset breakdown

Language averages

Rank	Model	Average WER	Coverage	RTFx	Size (B)	DE German	FR French	IT Italian	ES Spanish	PT Portuguese
1	microsoft/azure-speech-06-2026	2.61	5/5	NA	NA	1.9	3.27	2.51	2.25	3.1
2	elevenlabs/scribe_v2	2.65	5/5	NA	NA	2.24	3.29	2.49	2.32	2.9
3	reson8/resonant-1	3.2	5/5	NA	NA	2.78	4.11	2.96	2.66	3.5
4	reson8/resonant-1-flash	3.2	5/5	NA	NA	2.78	4.11	2.96	2.66	3.5
5	assemblyai/universal-3-pro	3.37	5/5	NA	NA	2.59	3.82	4.1	2.49	3.86
6	mistralai/Voxtral-Small-24B-2507	3.61	5/5	98.83	24	2.9	4.11	3.75	2.91	4.36
7	modulate/multilingual	3.72	5/5	NA	NA	3.44	4.14	3.16	2.98	4.9
8	CohereLabs/cohere-transcribe-03-2026	3.78	5/5	647.75	2	3.1	4.02	2.97	2.81	5.99
9	speechmatics/enhanced	4.16	5/5	NA	NA	2.78	4.89	5.43	2.74	4.97
10	soniox/stt-async-v5	4.29	5/5	NA	NA	3.64	6.21	4.2	3.37	4.01
11	facebook/omniASR-LLM-7B-v2	4.36	5/5	45.32	7.8	4.39	5.25	3.61	3.38	5.18
12	microsoft/Phi-4-multimodal-instruct	4.48	5/5	125.36	6	4.12	5.04	4.29	3.79	5.16

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Poradie	Model	Kategória	WER ▲	Vloženia	Vynechania	Zámeny	CER	DSER	RTFx
1	Google Chirp 3	commercial API	10.79%	2.63%	1.61%	6.56%	4.88%	19.14%	–
2	ElevenLabs Scribe v2	commercial API	11.93%	3.24%	1.23%	7.45%	5.52%	25.99%	–
3	OpenAI GPT Realtime Whisper	commercial API	14.18%	2.04%	2.71%	9.43%	6.18%	32.84%	–
4	Nvidia Canary 1B v2	open-weight	17.48%	1.67%	4.24%	11.57%	7.54%	23.08%	96.8x
5	Whisper Large v3	open-weight	18.72%	2.16%	4.18%	12.38%	8.47%	25.21%	109.5x
6	Whisper Large v3 Turbo	open-weight	20.71%	2.57%	4.90%	13.24%	9.69%	26.22%	313.9x
7	OpenAI GPT-4o Transcribe	commercial API	24.45%	1.04%	13.27%	10.14%	15.67%	29.59%	–
8	Omni ASR LLM 7B	open-weight	26.15%	4.60%	9.90%	11.65%	17.24%	48.32%	7.5x
9	Nvidia Parakeet TDT 0.6B v3	open-weight	27.20%	1.95%	6.13%	19.11%	11.38%	33.63%	470.5x
10	Omni ASR CTC 3B	open-weight	28.24%	1.10%	14.04%	13.10%	16.80%	74.70%	312.5x
11	Omni ASR CTC 7B	open-weight	28.47%	1.19%	14.96%	12.33%	17.70%	75.63%	169.6x
12	Omni ASR LLM 3B	open-weight	29.73%	4.12%	12.72%	12.89%	19.74%	71.44%	9.9x
13	Omni ASR CTC 1B	open-weight	30.48%	1.26%	15.12%	14.10%	17.92%	73.80%	520.4x
14	Microsoft VibeVoice ASR	open-weight	30.79%	4.76%	3.65%	22.37%	12.84%	32.71%	20.7x
15	Omni ASR LLM 1B	open-weight	31.00%	3.94%	14.55%	12.51%	21.26%	69.74%	12.4x
16	Whisper Medium	open-weight	31.47%	3.88%	5.92%	21.67%	13.54%	33.26%	149.1x
17	Omni ASR LLM 300M	open-weight	33.21%	4.10%	12.81%	16.30%	19.79%	72.26%	13.5x
18	Omni ASR CTC 300M	open-weight	40.16%	1.73%	15.64%	22.80%	19.06%	80.69%	690.3x
19	ESPnet OWSM CTC v4 1B	open-weight	42.96%	3.07%	7.72%	32.17%	15.36%	61.16%	146.0x
20	Whisper Small	open-weight	49.75%	7.32%	6.91%	35.51%	19.86%	39.81%	230.0x
21	Whisper Base	open-weight	76.71%	10.91%	8.06%	57.74%	31.71%	107.29%	323.7x
22	Whisper Tiny	open-weight	98.50%	21.28%	7.63%	69.59%	47.17%	267.06%	308.6x

SOTA STT model families

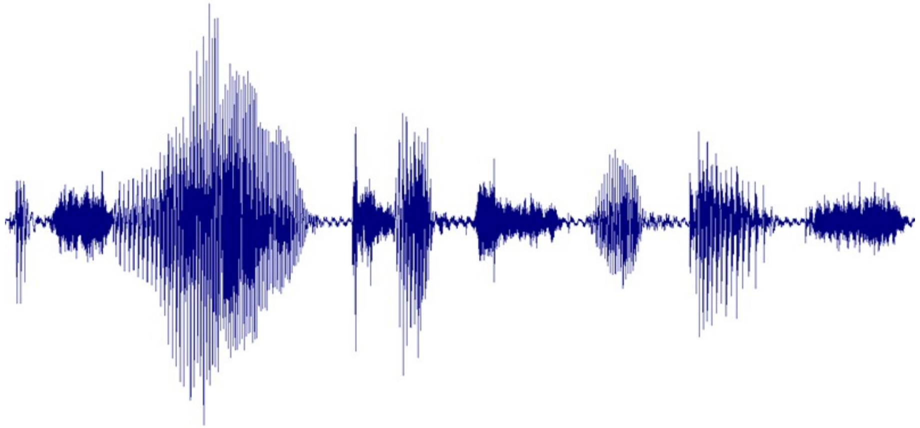
Family	Description	Examples
CTC encoder	Encoder + linear classification with CTC loss. Fast non-autoregressive inference.	Nvidia Parakeet-CTC, Meta OmniASR-CTC
RNN-Transducer / TDT	Encoder + lightweight prediction network build for streaming.	Nvidia Parakeet-TDT, Nvidia Nemotron ASR
Transformer encoder-decoder	Both encoder and decoder are transformers trained together end-to-end.	**OpenAI Whisper**, Nvidia Canary
Speech encoder + LLM	Encoder feeds an LLM that generates the transcript. Slower, but top performance.	Qwen-ASR, Nvidia Canary-Qwen, IBM Granite-Speech, Mistral Voxtral, OmniASR-LLM
Closed proprietary APIs	Likely a large transformer with LLM but we do not know.	Azure Speech, ElevenLabs Scribe, Google Chirp

Whisper

arxiv.org/abs/2212.04356

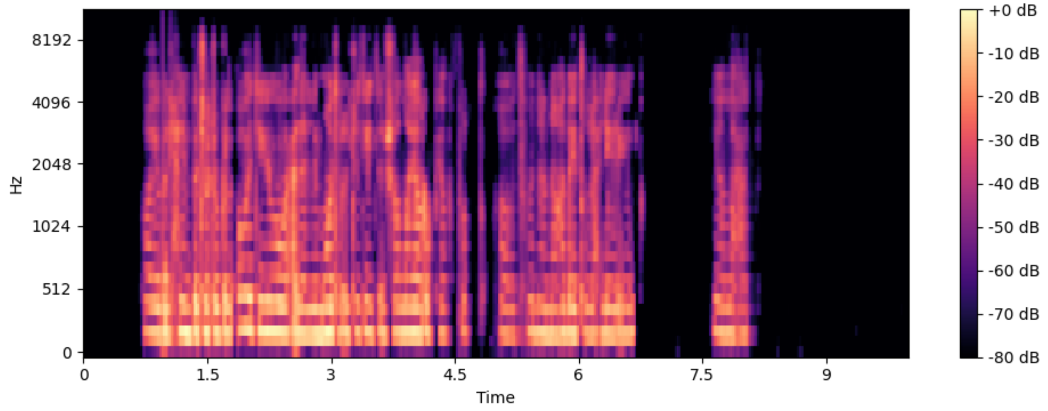
Audio as model input

- Sound is a pressure deviation propagating through a medium
- By sampling and quantization, we get a **digital waveform** (array)



Audio as model input

- Sound is a pressure deviation propagating through a medium
- By sampling and quantization, we get a digital waveform (array)
- We transform waveform to **Log-mel spectrogram** (matrix)
  - Created to represent human speech
  - Energy per frequency and time
  - Short-time Fourier transform
  - Mel filter bank
  - Log compression



First large-scale ASR training

- Whisper is a broadly used Speech-to-text model from OpenAI
- First release in 2022, last version in October 2024
- Common encoder-decoder transformer architecture with minimal change to support Automatic Speech Recognition

Train data matter a lot

- Amount of pre-training data per language is **highly predictive of zero-shot performance.**
- Only 90h of Slovak in the first version (0.013%)

we are here

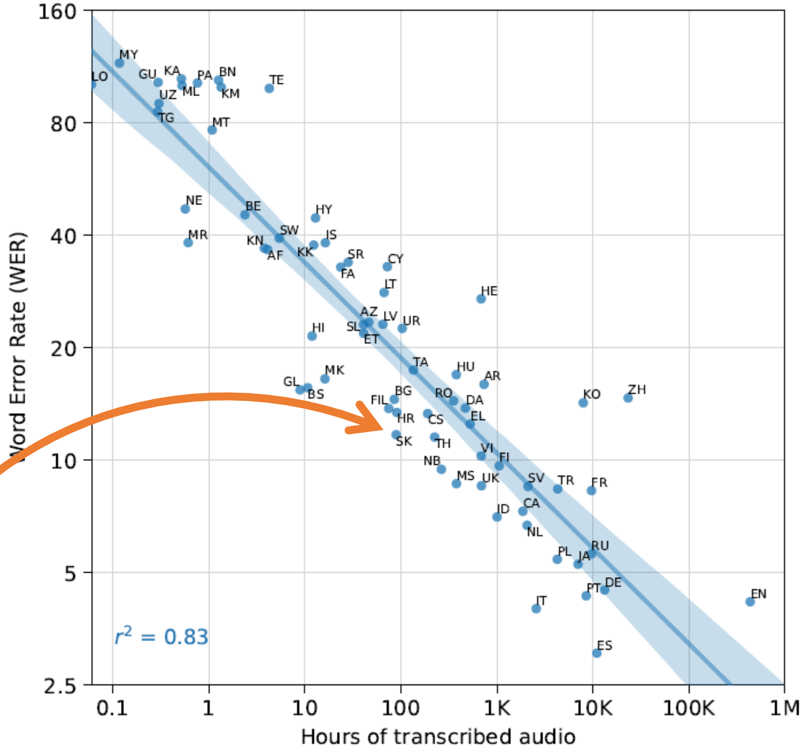
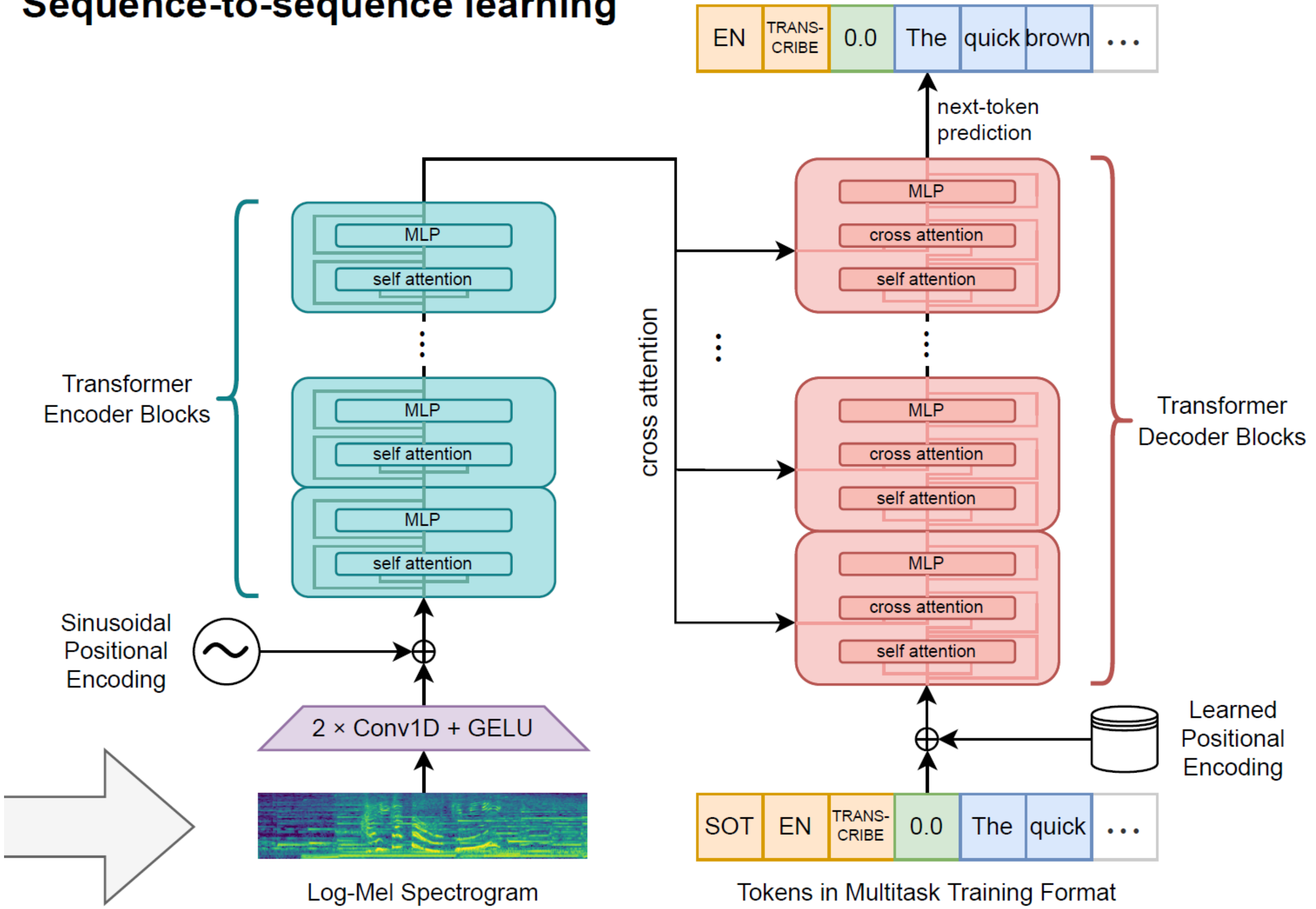


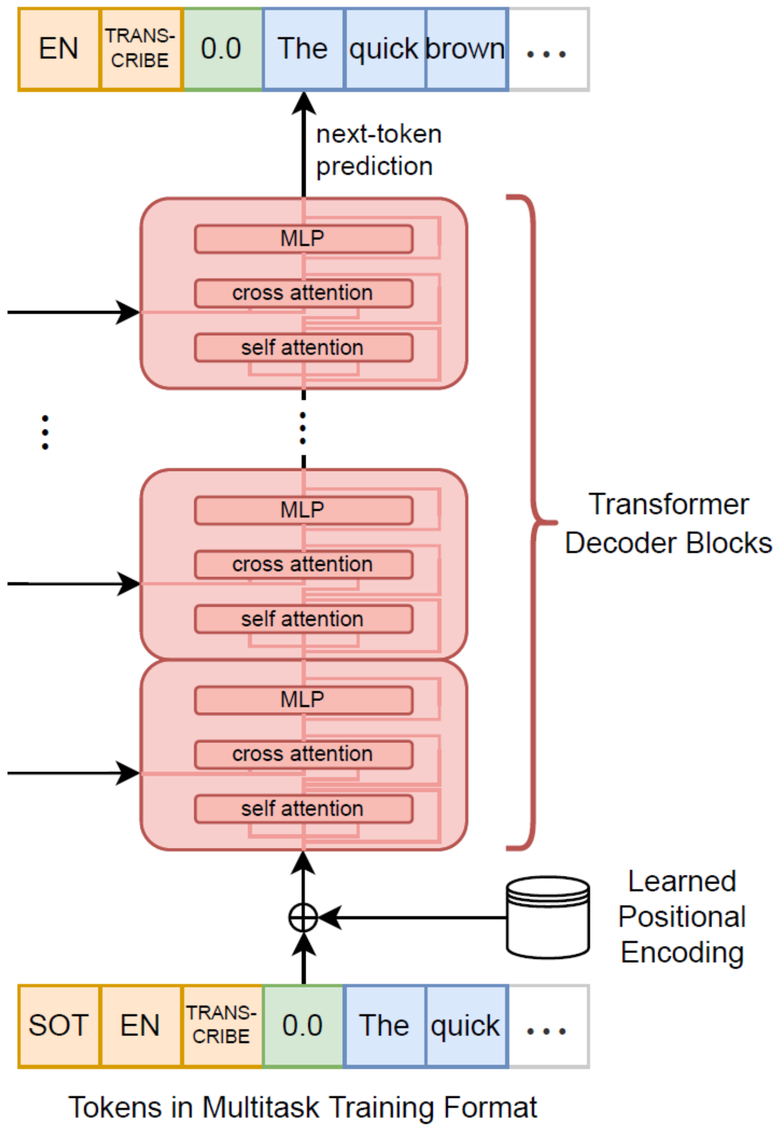
Figure 3. **Correlation of pre-training supervision amount with downstream speech recognition performance.** The amount of pre-training speech recognition data for a given language is very predictive of zero-shot performance on that language in Fleurs.

Architecture

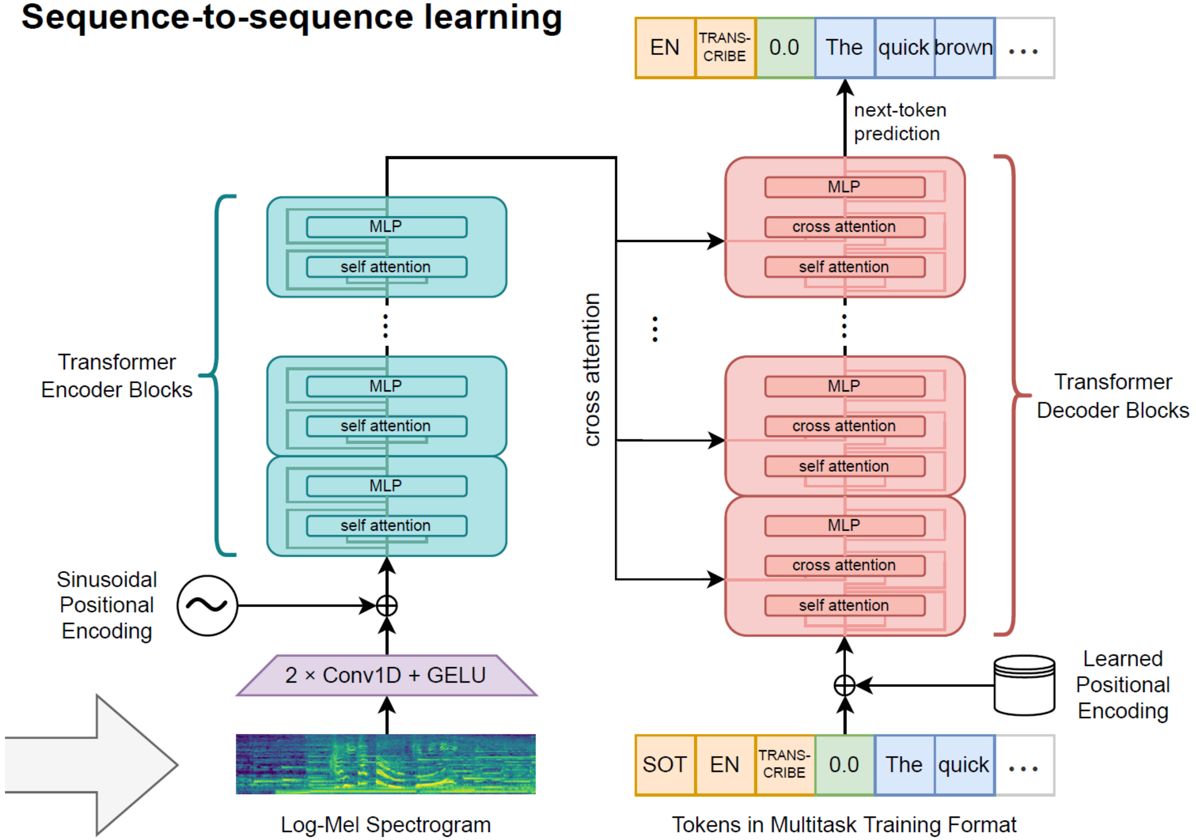
Sequence-to-sequence learning



This is basically your standard decoder-only LLM



Sequence-to-sequence learning

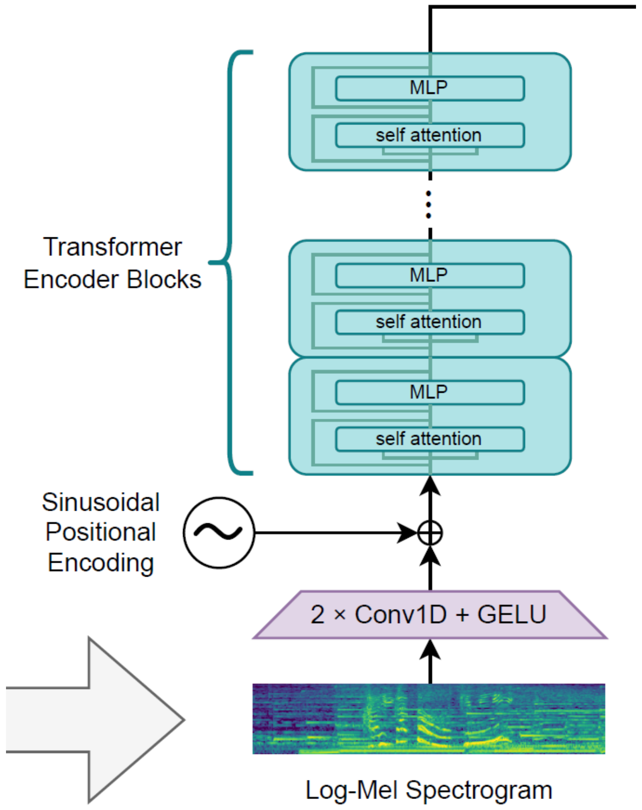


Model	Layers	Width	Heads	Parameters
Tiny	4	384	6	39M
Base	6	512	8	74M
Small	12	768	12	244M
Medium	24	1024	16	769M
Large	32	1280	20	1550M

Table 1. Architecture details of the Whisper model family.

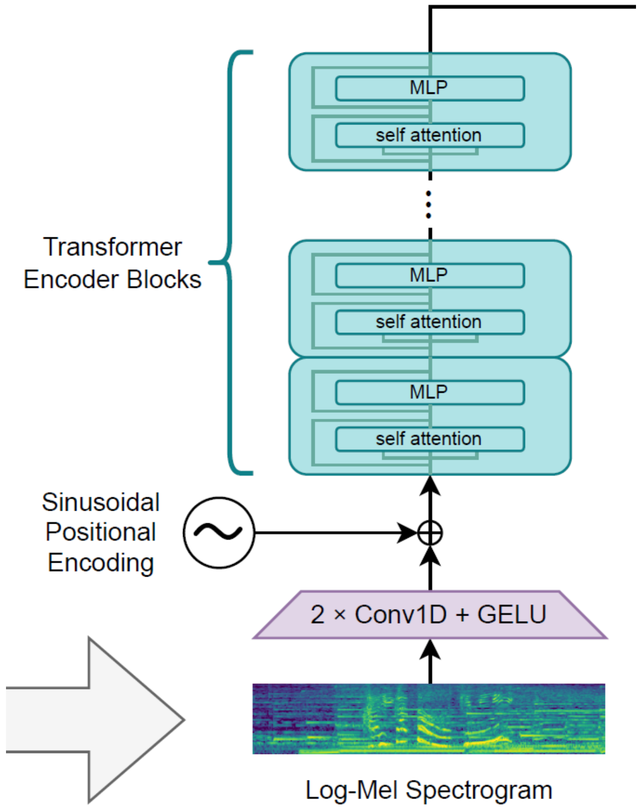
Whisper: Input

- Expects 30 seconds of audio at 16 kHz sampling rate
- Transformed to 80-channel log-Mel spectrogram
- **Conv1D #1**
  - maps channels to model dimensions
  - Extracts local temporal features (mixes channels across 3 frames)
- **Conv1D #2**
  - halves time frames (stride 2): 3000×1280 → 1500×1280
  - Cuts attention compute cost by 4×
- Fixed sinusoidal positional encoding added to frame embeddings



Whisper: Encoder Blocks

- Each block applies pre-norm residuals:
  - `x = x + MHSA(LayerNorm(x))`
  - `x = x + FFN(LayerNorm(x))`
- **FFN:** `Linear(d, 4d) → GELU → Linear(4d, d)`
- **Output:** final LayerNorm
- full self-attention, no mask
- 4 to 32 layers depending on model size



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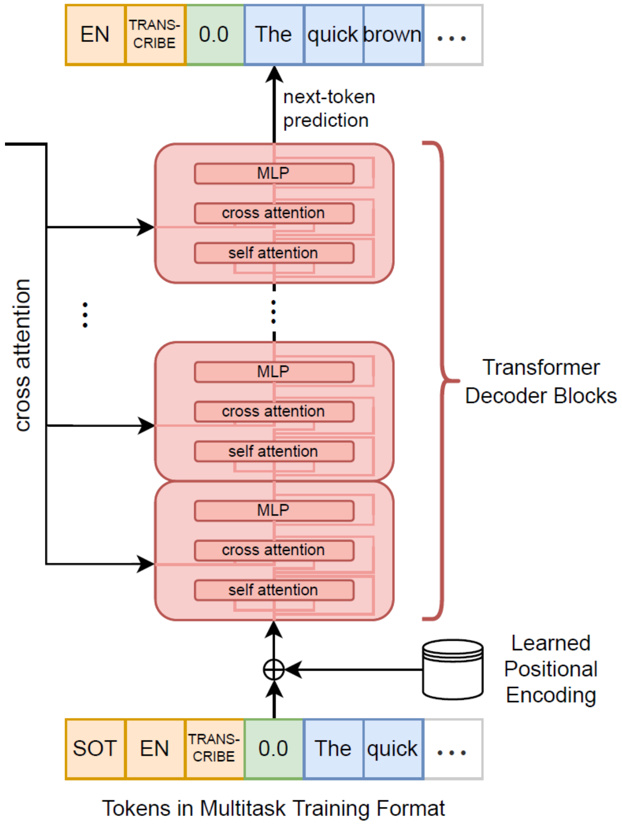
Whisper: Decoder

- Learned positional encoding
- Each block has 3 standard sub-layers:
  - `x = x + MHSA(LayerNorm(x))` with causal mask
  - `x = x + CrossAttention(LayerNorm(x))`
  - `x = x + FFN(LayerNorm(x))`
- Final LayerNorm after all blocks, then logits via transposed embedding matrix
- Same number of transformer layers as encoder (4 - 32)



Whisper: Decoder

KV cache

- Encoder K, V computed once and stored
- Decoder reuses past K, V from previous positions (standard LLM pattern)

Weight tying

- Same embedding matrix used for input tokens and output logits
- Biggest single matrix in the model: $|V| \times d \approx 66.4$M params (in large model)

Learned Positional encoding

- following GPT-2 convention unlike the encoder’s sinusoidal encoding

Whisper: Output

Logit filters enforce structure at inference

- Suppress control tokens during text generation
- Timestamps must occur in pairs and be monotonically non-decreasing

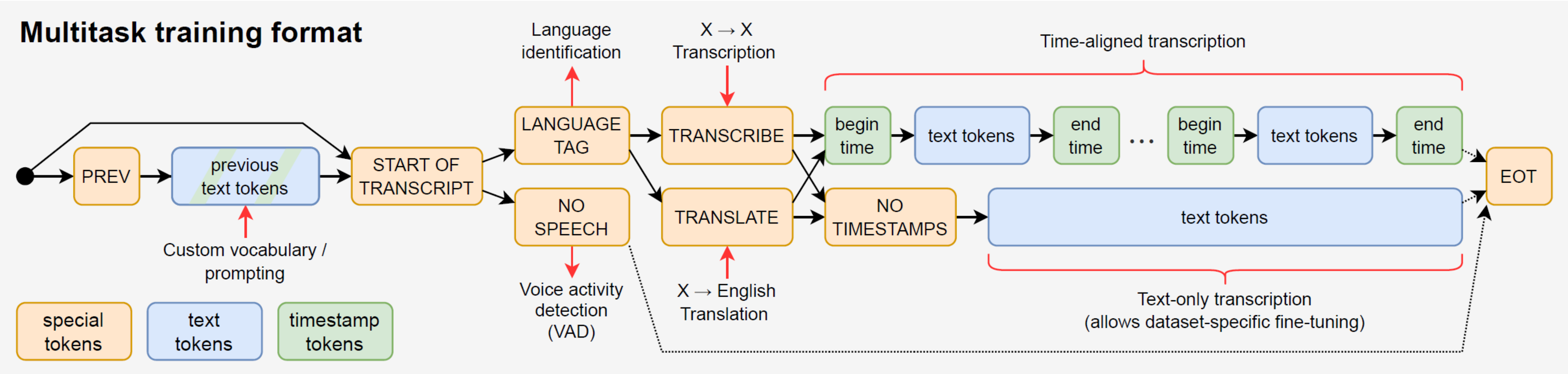
Long-form transcription strategy

- Sliding window over 30s segments, shifted by predicted timestamps
- 5-beam search with 0 temperature and fallback
  - increase temp by 0.2 (up to 1) if avg. log-prob < -1 or gzip compression rate > 2.4
- **No speech:** skip segment if P(nospeech) > 0.6 and avg. log-prob < -1

Training & Evaluation

Whisper: Multitask Training Format

All tasks encoded as a **single sequence of special tokens** so one model replaces many pipeline stages.



Whisper: Multitask Training Format

Special tokens (not all):

- `<|startoftranscript|>`
- `<|nospeech|>`
- `<|transcribe|>`
- `<|translate|>`
- `<|notimestamps|>`
- `<|endoftranscript|>`
- 1501 special time tokens (30 seconds by 20ms steps):
  - `<|0.00|>, <|0.02|>, ..., <|29.98|>, <|30.00|>`
- language tokens: `"en", "zh", ...`

Whisper: Multitask Training Format

- **Tasks learned jointly:** transcription (any language), translation (X → en), language detection, voice activity detection, timestamping
- The initial tokens are pre-filled during inference, not generated
- Language detected via a separate forward pass if not specified
  - greedy pick over language tokens only
- All control tokens are suppressed during transcript generation
- Multitask helps larger models (positive transfer), but hurts smaller ones

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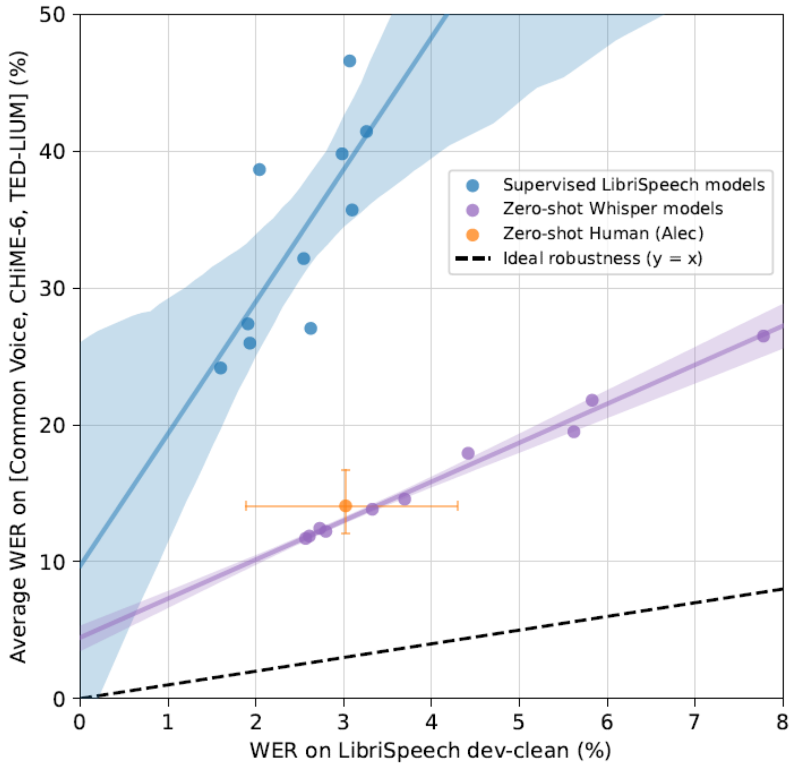


Figure 2. **Zero-shot Whisper models close the gap to human robustness.** Despite matching or outperforming a human on LibriSpeech dev-clean, supervised LibriSpeech models make roughly twice as many errors as a human on other datasets demonstrating their brittleness and lack of robustness. The estimated robustness frontier of zero-shot Whisper models, however, includes the 95% confidence interval for this particular human.

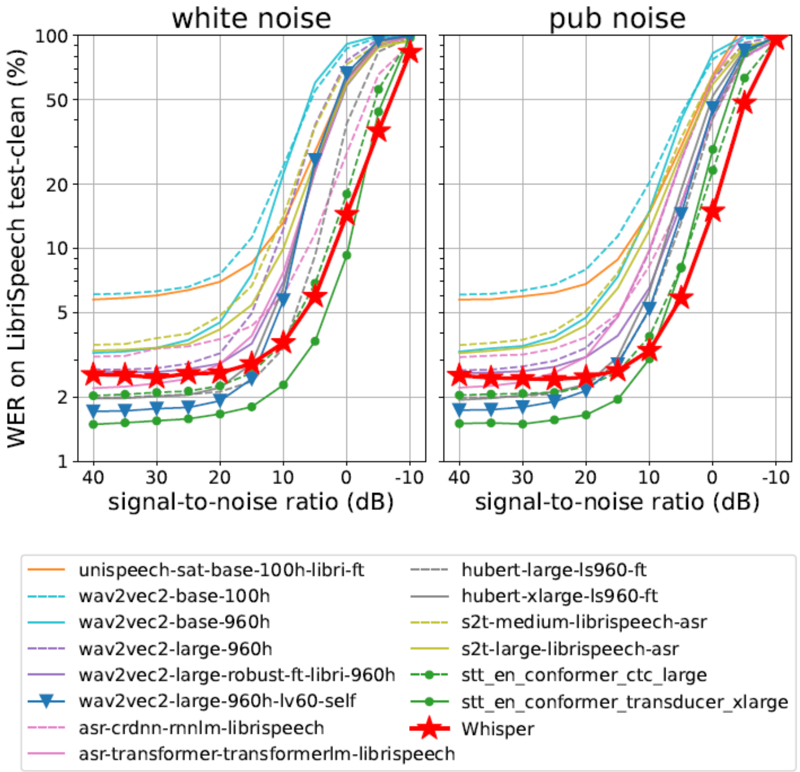


Figure 5. **WER on LibriSpeech test-clean as a function of SNR under additive white noise (left) and pub noise (right).** The accuracy of LibriSpeech-trained models degrade faster than the best Whisper model (★). NVIDIA STT models (●) perform best under low noise but are outperformed by Whisper under high noise (SNR $<$ 10 dB). The second-best model under low noise (▼) is fine-tuned on LibriSpeech only and degrades even more quickly.

Demo Notebooks

What are the capabilities today

- Commercial services are close to realistic voice generation
- English ASR is super-human in many cases
- What about open-source?
- Huge difference between languages
  - English / Chinese
  - Big languages (e.g., DE, FR, ES)
  - Low-resource languages
  - Previously completely unsupported
- Most languages do not even have a benchmark

Back to the demo notebooks

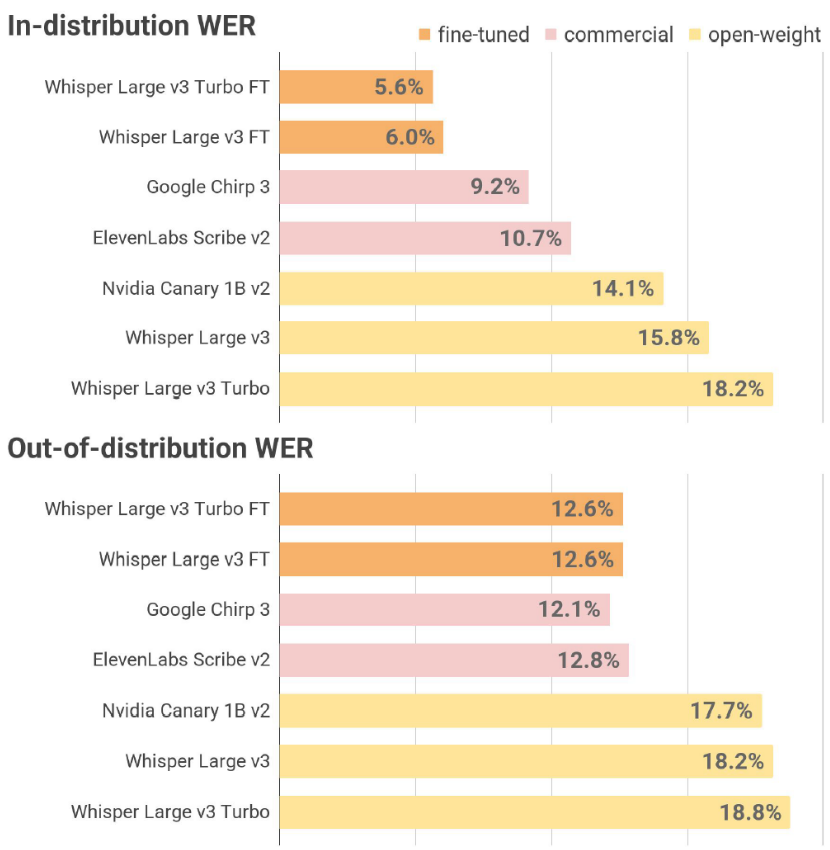
- Speech-to-text Colab Demo
- Text-to-Speech Colab Demo

tinyurl.com/kinitnlp

Fine-tuning ASR in practice

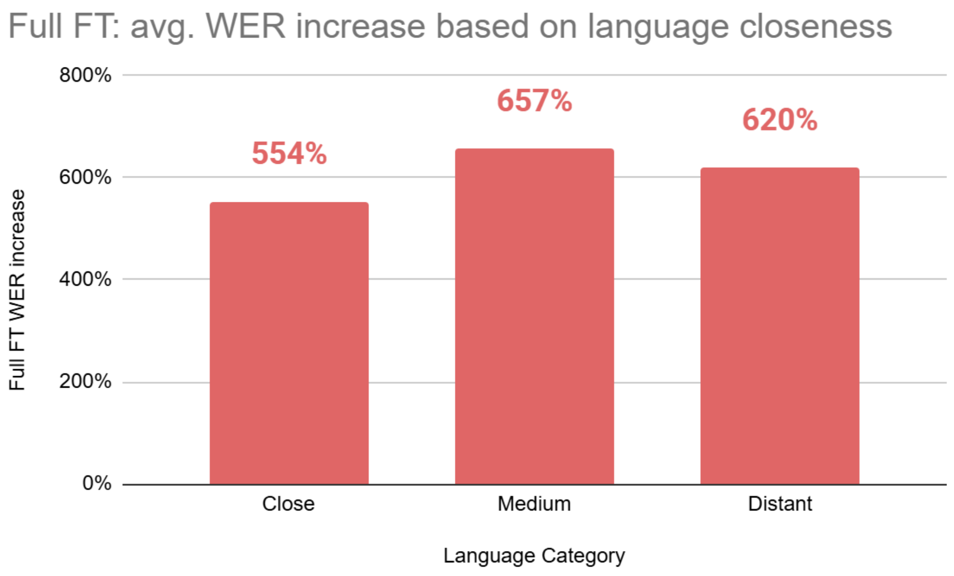
Fine-tuning works with limitations

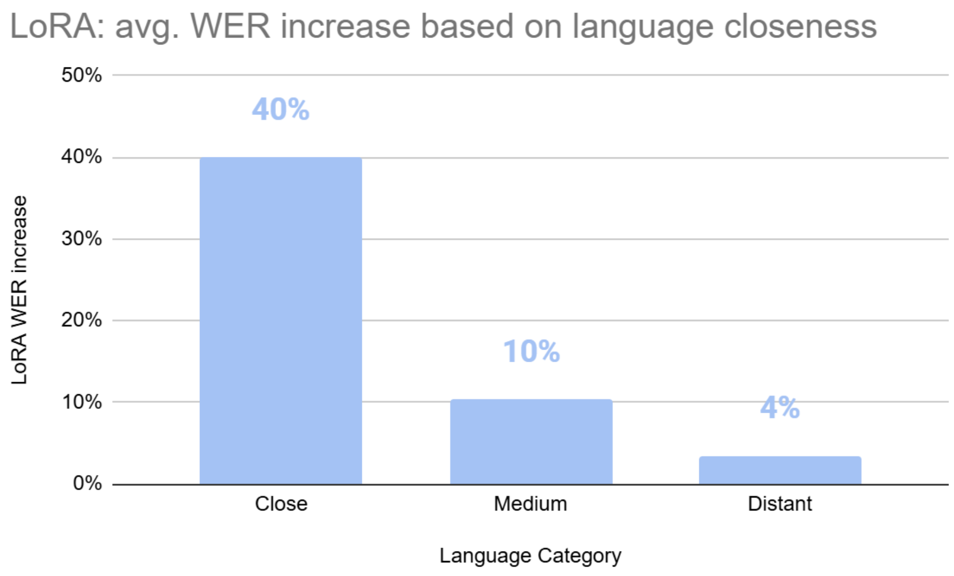
Fine-tuning works well in domain, but the improvement does not necessarily generalize out of domain.



Languages are forgotten

Catastrophic forgetting is brutal for languages excluded from full fine-tuning.





Data Sources

What is the issue with low-resource languages?

- Data
- Harder to get audio than text
- Even harder to get annotated audio
  - You can train with weak labels
  - But you really need human annotation for evaluation
- Legal constraints for publishing
  - Speech as a personal identifier
  - GDPR

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Tips to get audio data with reference labels

- Public recordings with human transcription
  - SloPalSpeech
  - Municipalities
  - Bible or other old books with public recordings
  - Needs non-trivial processing pipeline (transcription + forced alignment)
- Scientific work with published datasets
  - TEDxSK and JumpSK Lecture Speech Corpus
- Hugging Face and Mozilla Common Voice

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